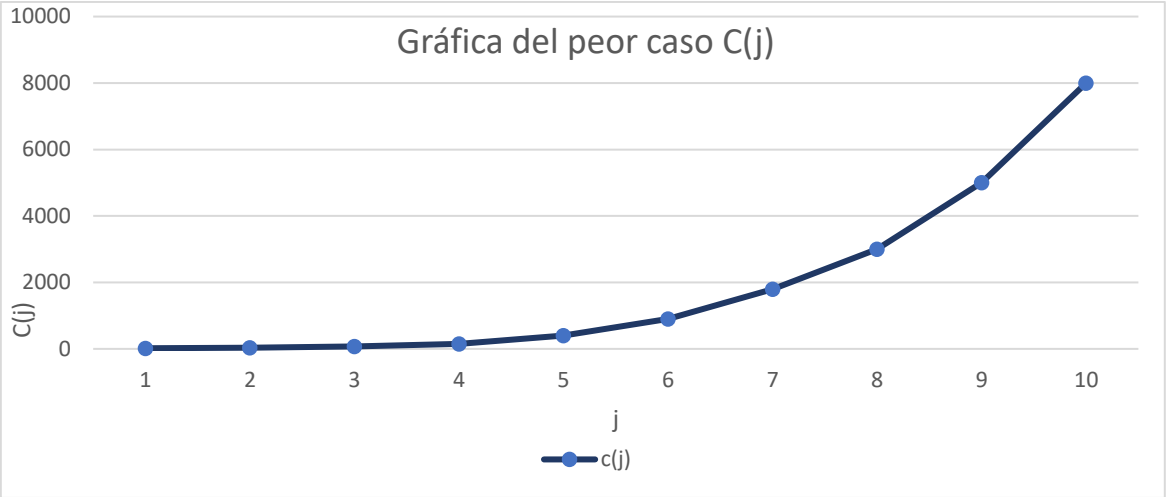
Ordenar las funciones

1. $\log\log n$
2. $\log n$,
3. $\ln n$
4. $(\log n)^2$
5. $n^{\frac{1}{2}}$
6. n
7. $n\log n$
8. $n^{1+\varepsilon}$
9. $n^2, n^2 + \log n$
10. n^3
11. $n - n^3 + 7n^5$
12. 2^{n-1}
13. 2^n
14. e^n
15. $n!$

Ejecución big oh

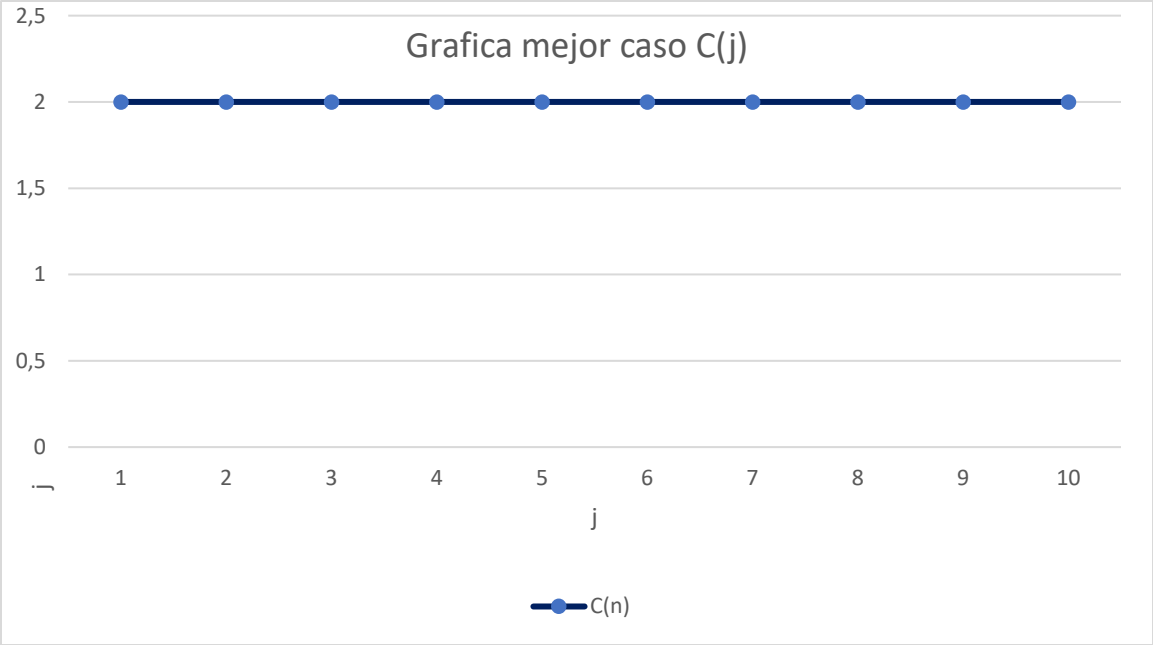
	costo	tiempo
$function\ mystery(n)$		
$r \coloneqq 0$	C1	1
$for\ i \coloneqq 1\ to\ n-1\ do$	C2	n-1
$for\ j \coloneqq i+1\ to\ n\ do$	C3	n(n-2)
$for\ k \coloneqq 1\ to\ j\ do$	C4	(n-2) (n-1) ($\sum_{k=1}^{j} t_k$)
$r \coloneqq r+1$	C5	(n-2) (n-1) ($\sum_{k=1}^{j} t_k - 1$)
return(r)	C6	1

$$\text{Peor C} = \text{C1} + \text{C2}(\text{j} - 1) + \text{C3}(j^2 - 2\text{j}) + \text{C4}\left(\frac{1}{2}j^4 - j^3 - \frac{1}{2}j^2 + \text{j}\right)$$
$$+ \text{C5}\left(\frac{1}{2}j^4 - j^3 - \frac{3}{2}j^2 + 4\text{j} - 2\right) + \text{C6}$$



La gráfica muestra el crecimiento del costo en el peor caso.Dado que el crecimiento es cúbico, el algoritmo pertenece a la clase de complejidad **O(n³)**.

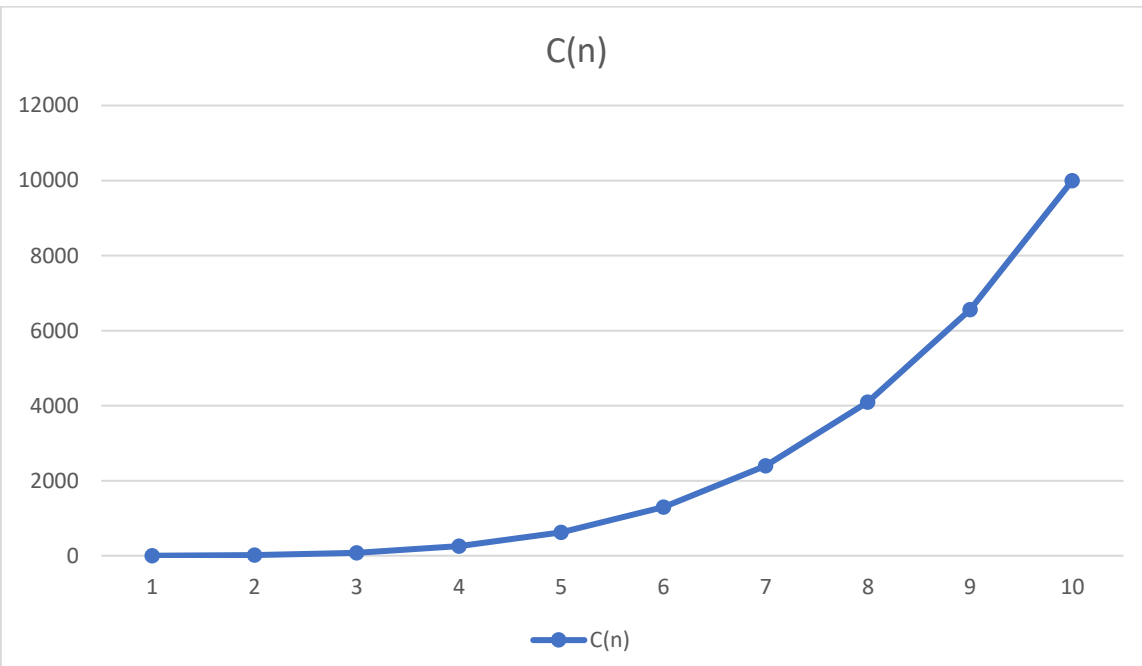
$$\text{Mejor } \ C = C1 + C6$$



$$\text{Big Oh} = \ O(n^4)$$

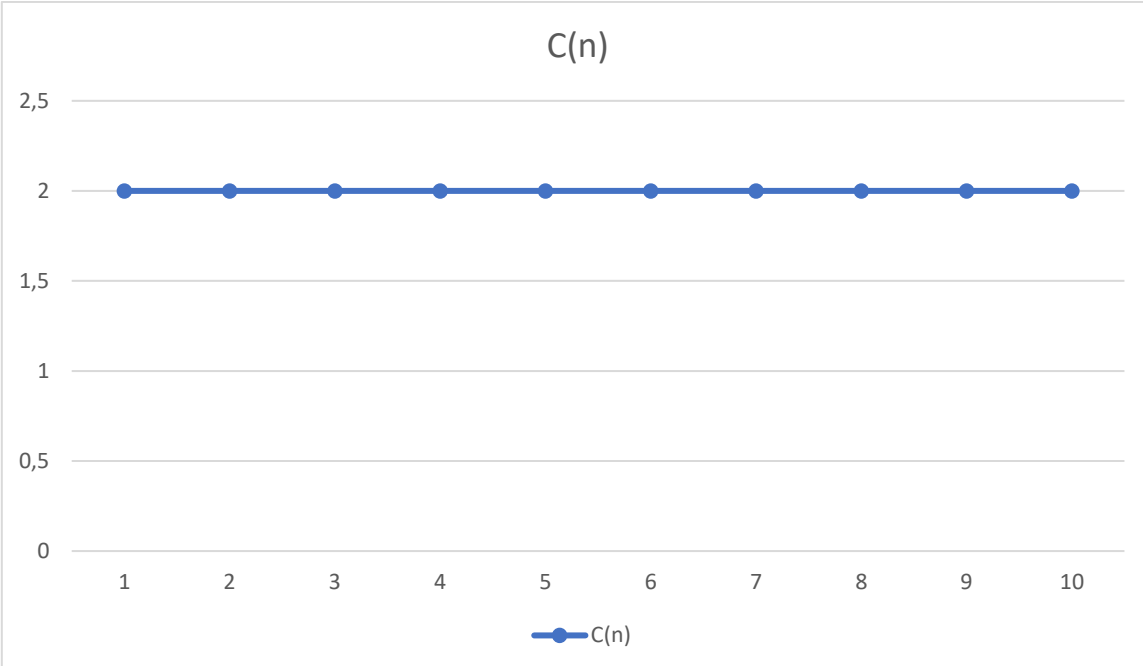
$function\ pesky(n)$		
$r \coloneqq 0$	C1	1
$for\ i \coloneqq 1\ to\ n\ do$	C2	n
$for\ j \coloneqq 1 to\ i\ do$	C3	(n-1) $(\sum_{j=1}^{i} t_j)$
$for\ k \coloneqq j\ toi +\ j\ do$	C4	(n-1) $(\sum_{j=1}^{i} t_j - 1)(\sum_{k=j}^{i+j} t_k)$
$r \coloneqq r + 1$	C5	(n-1) $(\sum_{j=1}^{i} t_j - 1)(\sum_{k=j}^{i+j} t_k - 1)$
return(r)	C6	1

$$\text{Peor C} = \text{C1} + \text{C2(j)} + \text{C3}\left(\frac{1}{2}j^3 - \frac{1}{2}\text{j}\right) + \text{C4}\left(\frac{1}{4}j^5 + \frac{1}{4}j^4 - \frac{3}{4}j^3 - \frac{1}{4}j^2 + \frac{1}{2}\text{j}\right)$$
$$+ \text{C5}\left(\frac{1}{8}j^7 - \frac{1}{2}j^5 + \frac{7}{8}j^3 + \frac{1}{4}j^6 - \frac{3}{4}j^4 - \frac{1}{2}j + \frac{1}{2}j^2\right)$$
$$+ \text{C6}\left(\frac{1}{8}j^7 - j^4 - \frac{5}{2}j + \frac{1}{4}j^6 - \frac{3}{4}j^5 + \frac{17}{8}j^3 + \frac{3}{4}j^2 + 1\right) + \text{C7}$$



a gráfica representa el crecimiento del costo computacional en el peor caso.
Dado que el crecimiento es proporcional a n^4, el algoritmo pertenece a la clase de complejidad **$O(n^4)$**.

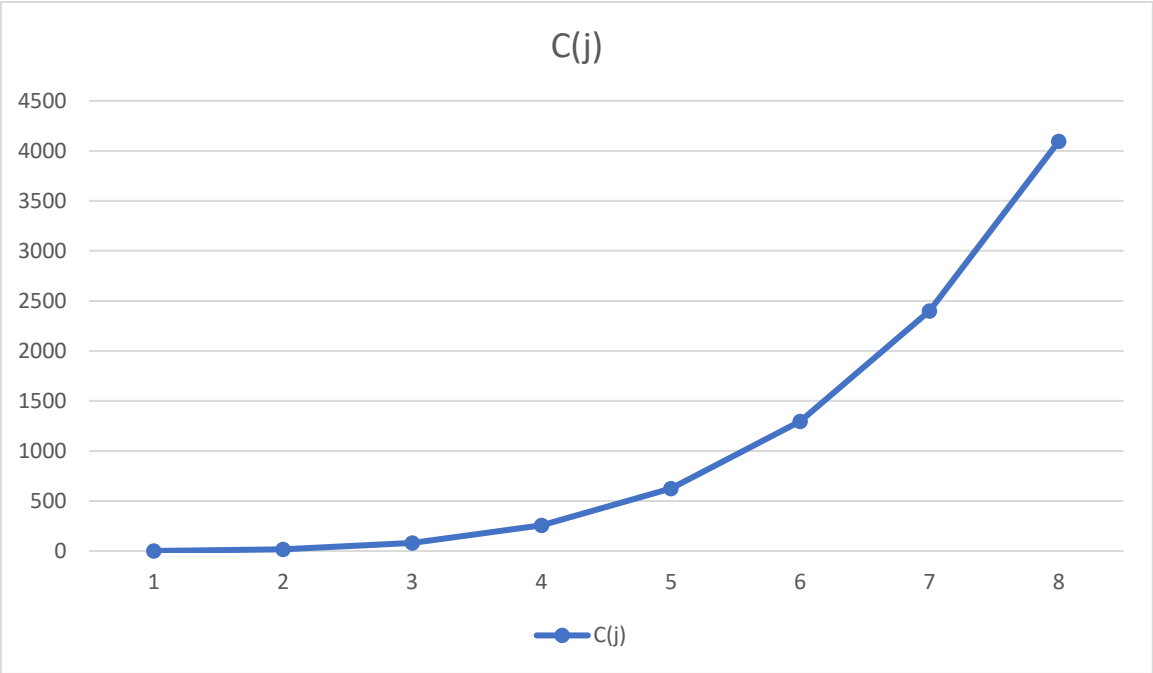
$$\text{Mejor C} = \text{C1} + \text{C6}$$



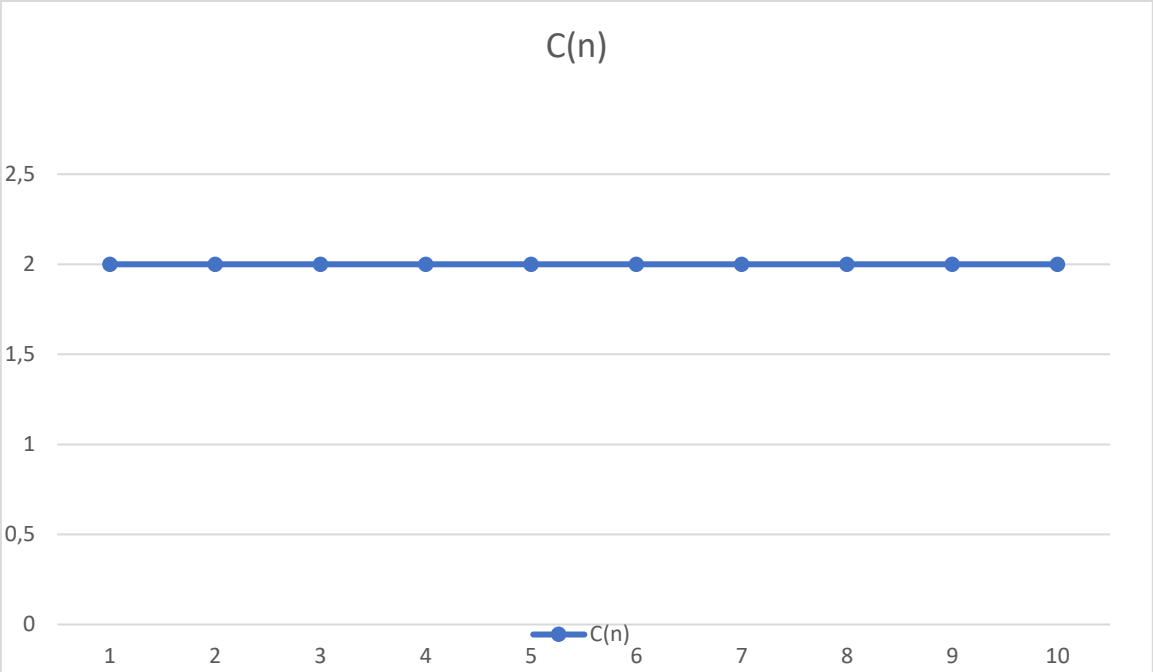
Big Oh = $O(n^5)$

$function\ prestiferous(n)$		
$r := 0$	C1	1
$for\ i := 1\ to\ n\ do$	C2	n
$for\ j := 1\ to\ i\ do$	C3	(n-1) $(\sum_{j=1}^{i} t_j)$
$for\ k := j\ to\ i + j\ do$	C4	(n-1) $(\sum_{j=1}^{i} t_j - 1)(\sum_{k=j}^{i+j} t_k)$
$for\ l := 1\ to\ i + j - k\ do$	C5	(n-1) $(\sum_{j=1}^{i} t_j - 1)(\sum_{k=j}^{i+j} t_k - 1)\ (\sum_{L=1}^{i+j-k} t_L)$
$r := r + 1$	C5	(n-1) $(\sum_{j=1}^{i} t_j - 1)(\sum_{k=j}^{i+j} t_k - 1)\ (\sum_{L=1}^{i+j-k} t_L - 1)$
return(r)	C6	1

$$\text{Peor C} = \text{C1} + \text{C2(j)} + \text{C3}\left(\frac{1}{2}j^3 - \frac{1}{2}\text{j}\right) + \text{C4}\left(\frac{1}{4}j^5 + \frac{1}{4}j^4 - \frac{3}{4}j^3 - \frac{1}{4}j^2 + \frac{1}{2}\text{j}\right)$$
$$+ \text{C5}\left(\frac{1}{8}j^7 - \frac{1}{2}j^5 + \frac{7}{8}j^3 + \frac{1}{4}j^6 - \frac{3}{4}j^4 - \frac{1}{2}j + \frac{1}{2}j^2\right)$$
$$+ \text{C6}\left(\frac{1}{8}j^7 - j^4 - \frac{5}{2}j + \frac{1}{4}j^6 - \frac{3}{4}j^5 + \frac{17}{8}j^3 + \frac{3}{4}j^2 + 1\right) + \text{C7}$$



Mejor $C = C1 + C7$



Big Oh $= O(n^7)$